

Akash Sonth

Boston, MA, USA | akashsonth@gmail.com | +1 540-355-6787 | akashsonth.github.io

EDUCATION

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| Virginia Polytechnic Institute and State University (Virginia Tech)
<i>M.S., Computer Engineering (concentration: Machine Learning), GPA: 3.88/4</i>
• Thesis: Enhancing road safety through machine learning for prediction of unsafe driving behaviors | Blacksburg, VA, USA
<i>Aug 2021 - Jul 2023</i> |
| Birla Institute of Technology and Science, Pilani (BITS Pilani)
<i>B.E. (Hons.), Electrical and Electronics Engineering, GPA: 8.39/10</i>
• Thesis (at Brno University of Technology, Czechia): Vehicle density prediction and counting from monocular images
• Member, SAE Formula Student team: End-to-end design and integration of a Formula Student race car. | Pilani, India
<i>Aug 2015 - Jul 2019</i> |

TECHNICAL SKILLS

Programming	Python, C#, MATLAB, SQL, $\text{\LaTeX}$
Frameworks	PyTorch, PyG, Keras, TensorFlow, OpenCV, Open3D, scikit-learn
Robotics & Simulation	ROS, OpenStreetMap (OSM), Unity, Raspberry Pi
Systems & Cloud	Git, Docker, Linux, Bash, CI/CD, AWS (EC2, S3)
Agentic AI	Microsoft Agent Framework

EXPERIENCE

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| Emerson (formerly Aspen Technology)
<i>Data Scientist</i>
• Developed an agentic framework for outlier detection, data quality assessment, and failure-relevant sensor selection in industrial time-series datasets, reducing analysis time from 2 hrs to under 5 mins per asset. (<i>Patent pending</i>)
• Designed an unsupervised equipment monitoring algorithm fusing expert-defined and generative-model-derived signatures, improving detection lead time by up to 10 hrs over production methods. (<i>Patent pending</i>) | Bedford, MA, USA
<i>Aug 2023 - Present</i> |
| Virginia Tech Transportation Institute
<i>Graduate Research Assistant</i>
• Designed a multi-stage perception pipeline to model traffic scenes as semantic scene graphs using Lanelet2 maps in OpenStreetMap format for lane-level topology.
• Developed TransformerConv and GINEConv based graph neural networks for intersection safety prediction, achieving 79% accuracy on low-resolution (320×240, 15 fps) videos.
• Adapted the Video Swin Transformer for explainable driver distraction monitoring by incorporating a visual dictionary of human actions, achieving 80.2% Top-1 accuracy on a large-scale low-quality dataset of 10k+ videos.
• Built a containerized Model Zoo for deployment of eye gaze estimation, multi-object tracking, and panoptic segmentation models. | Blacksburg, VA, USA
<i>Dec 2021 - Aug 2023</i> |
| Visual Computing Lab, Indian Institute of Science
<i>Research Assistant</i>
• Proposed an identity-free pose transformation method using non-parametric template cropping. Achieved state-of-the-art 93.6% Rank-1 accuracy on Market-1501 (32K+ images) using a two-stage self-supervised learning framework with a novel radial distance loss and ResNet-50 backbone.
• Implemented distributed multi-node, multi-GPU training using the NCCL backend, enabling 4x larger batch-sizes. | Bengaluru, India
<i>Feb 2020 - Apr 2021</i> |

TEACHING EXPERIENCE

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| BITS Pilani F312 Neural Networks and Fuzzy Logic - Teaching Assistant | <i>Jan 2019 - May 2019</i> |
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PUBLICATIONS [[Google Scholar](#)]

- **Akash Sonth**, Lynn Abbott, Abhijit Sarkar. “*Intersection Safety Modeling Using Semantic Scene Graph and Graph Neural Network.*” IEEE Intelligent Vehicles Symposium (IV), 2025. [[Paper](#)][[Project Page](#)]
- **Akash Sonth**, Abhijit Sarkar, Hirva Bhagat, Lynn Abbott. “*Explainable Driver Activity Recognition Using Video Transformer in Highly Automated Vehicle.*” IEEE Intelligent Vehicles Symposium (IV), 2023. [[Paper](#)][[Project Page](#)]
- **Akash Sonth**, Harshavardhan Settibhaktini, Ankush Jahagirdar. “*Vehicle speed determination and license plate localization from monocular video streams.*” International Conference on Computer Vision & Image Processing (CVIP), 2018. [[Paper](#)]
- Siddharth Seth, **Akash Sonth**, Anirban Chakraborty. “*Pose-Transformation and Radial Distance Clustering for Unsupervised Person Re-identification.*” British Machine Vision Conference (BMVC), 2021. [[Paper](#)][[Project Page](#)]
- Hirva Bhagat, Sandesh Jain, Lynn Abbott, **Akash Sonth**, Abhijit Sarkar. “*Driver Gaze Fixation and Pattern Analysis in Safety Critical Events.*” IEEE Intelligent Vehicles Symposium (IV), 2023. [[PAPER](#)]

AWARDS & FELLOWSHIPS

- Oral Presentation, IEEE Intelligent Vehicles Symposium (IV) (Top 7% of accepted papers), 2025
- IEEE Intelligent Transportation Systems Society (ITSS) flagship conference travel grant, 2025
- Google Research India Summer School (Computer Vision track), 2020 - one of 50 students selected nationwide.
- Virginia Tech, Center for Human-Computer Interaction (CHCI) conference award, 2023
- Virginia Tech, Graduate and Professional Student Senate (GPSS) conference award, 2023

SELECTED TALKS

- “Intersection Safety Modeling Using Semantic Scene Graph and Graph Neural Network.” Invited talk at 2nd Workshop on Secure connected vehicles: Digital Twin, UAVs, and Smart Transportation, IEEE Intelligent Vehicles Symposium (IV), 2025.
- “Real-Time Risk Prediction at Signalized Intersection Using Graph Neural Network.” Virginia Tech Transportation Institute (VTTI), Safe-D webinar, 2023 [[Presentation](#)]